

UNIT - I

1

Introduction to Embedded Systems

Syllabus

Definition, Characteristics of Embedded System, Real time systems, Real time tasks. **Processor basics** : General Processors in Computer Vs Embedded Processors, Microcontrollers, Microcontroller Properties, Components of Microcontrollers, System-On-Chip and its examples, Components of Embedded Systems, Introduction to embedded processor.

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- 1.2 Real Time Systems
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- 1.5 System on Chip and its Example **May-19**Marks 4
- 1.6 Multiple Choice Questions

1.1 Definition of Embedded System

➡ SPPU : May-18

- A system is a set of interacting or interdependent component parts forming a complex unit. It is a fixed plan to perform one or many tasks.
- Embedded system is an electronic system which is designed to perform one or a limited set of functions using software and hardware.
- **General definition of embedded systems** is embedded systems are computing systems with tightly coupled hardware and software integration that are designed to perform a dedicated function. The word embedded reflects the fact that these systems are usually an integral part of a larger system, known as the embedding system. Multiple embedded systems can coexist in an embedding system.
- IEEE has the following definition for embedded systems : "An Embedded Computer System: A computer system that is part of a larger system and performs some of the requirements of that system; for example, a computer system used in an aircraft or rapid transit system".
- Embedded systems cover a large range of computer systems from ultra small computer-based devices to large systems monitoring and controlling complex processes.
- Fig. 1.1.1 shows general view of embedded system.

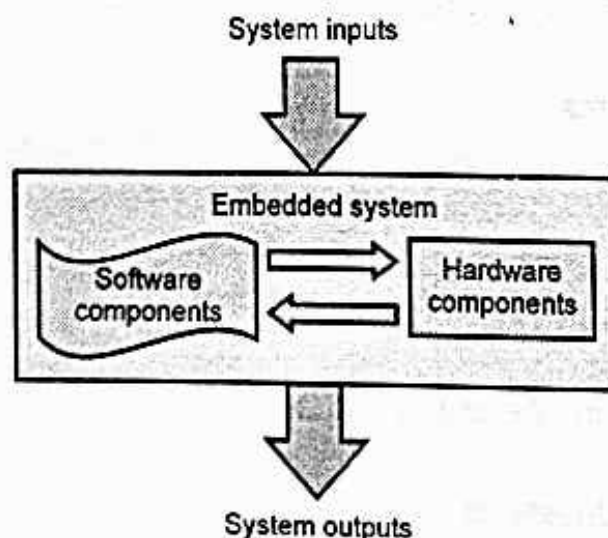


Fig. 1.1.1 : General view of embedded system

- Embedded systems are information processing systems that are embedded into a larger product and that are normally not directly visible to the user.

- In most of the cases embedded systems are real-time systems. In many cases embedded systems are safety or mission critical systems. Embedded systems vary from very small systems to very large systems. For small systems there are strong constraints related to different resources such as power or memory consumption.
- **Real-time systems** are those systems in which the overall correctness of the system depends on both the functional correctness and the timing correctness. Realtime systems also have a substantial knowledge of the system it controls and the applications running on it. These are deadline dependent.
- Embedded systems are not always standalone devices. Many embedded systems consist of small, computerized parts within a larger device that serves a more general purpose.

1.1.1 History

- Embedded systems were existing before the information technology arrives into the market. Initially vacuum tube used for designing embedded system. Programming for embedded system is done in low level programming language.
- Apollo guidance computer was the first modern embedded system. Developed by Charles Stark Draper at the MIT instrumentation laboratory. Lower price and rise in processing power and functionality was after the 1960.
- Autonetics D-17 was developed in 1961 for guidance computer for the Minuteman missile.
- The first microprocessor, the Intel 4004, was designed for calculators and other small systems but still required many external memory and support chips.
- Automobiles used microprocessor-based engine controllers in 1970's. It control fuel/air mixture, engine timing, etc. It also support multiple modes of operation : Warm-up, cruise, hill climbing, etc.
- First microcontroller TMS 1000 was developed in 1974 by Texas Instruments. It had RAM, ROM and clock circuitry on the chip along with the processing unit.
- In 1980, Intel introduced the 8051 MCU and called it MCS-51 architecture.
- Laser and inkjet printers also emerged in the 1980s. Print engines require computational support for both typesetting and real-time control.
- Cell phones use coprocessors for digital-signal processing and micro-processors for button control and protocol processing. By the early 1990s, a typical cell phone contained five or six DSPs and CPUs. The telephone infrastructure also uses embedded DSPs and microprocessors extensive.

1.1.2 Generation based Classification

Generation	Description	Example
First Generation (1G)	8-bit microprocessor and 4 bit microcontroller like 8085 and Z80 was used in first generation. - Hardware circuit is simple. - Assembly code is used for developing firmware.	- Digital telephone keypads - Stepper motor
Second Generation (2G)	- Uses 16-bit microprocessor and 8-bit microcontroller. - They are more complex and powerful than 1G microprocessor and microcontroller.	Data acquisition systems like ADC, SCADA system.
Third Generation (3G)	- Uses 32-bit microprocessor and 16-bit microcontroller. - Domain specific processor and controllers are used.	Robotics
Fourth Generation (4G)	- Uses 64-bit microprocessor and 32-bit microcontroller. - The concept of system on chips, multicore processors evolved. - Highly complex and very powerful.	Smart phones

1.1.3 Classification based on Performance and Complexity

- Embedded system is classified as :
 1. Small scale 2. Medium scale 3. Sophisticated esbedded System
- **Small scale** : Designed with a single 8 - or 16-bit microcontroller; little hardware and software complexities and involve board-level design. Tools for development of embedded software are editor, assembler and cross assembler, Integrated Development Environment (ISE). It uses battery operated power supply.
- **Medium scale** : These types of embedded systems design with a single or 16 bit or 32 bit microcontroller, RISCs or DSPs. These types of embedded systems have both hardware and software complexities. For developing embedded software for medium scale embedded systems, the main programming tools are C, C++, JAVA, Visual C++, RTOS, debugger, source code engineering tool, simulator and IDE.
- **Sophisticated embedded systems** : These types of embedded systems have massive hardware and software complexities that may need ASIPs, IPs, PLAs, scalable or configurable processors. They are used for flight landing gear systems, car braking systems, military applications and robots.

1.1.4 Characteristics of Embedded System

1. It has limited memory and limited number of peripherals.
 2. It should be designed for low power dissipation, because most of system uses battery power.
 3. They need to be highly reliable and stable.
 4. It should be small set of functions.
 5. Typically designed for a specific application or purpose
 6. They generally have minimal or no user interface.
 7. Embedded systems are typically designed to meet real time constraints
 8. The processor inside the embedded system should perform operations that are complex in nature
 9. Embedded systems are usually feedback oriented or reactive
- Industrial requirements depend on applications; special requirements typically include :
 1. Availability and reliability
 2. Safety

3. Survivability
4. Security
5. Real-time, deterministic response
6. Power consumption
7. Lifetime issues



1.1.5 Application Domain of Embedded System

- Embedded Systems were used for large, safety-critical and business-critical applications.

Embedded System Area	Examples
Automobiles and in telecommunications	• Car and motor control system • Engine safety • Entertainment and multimedia in car • E-Commerce • Robotics • Wireless communication • Mobile computing and networking
Smart Cards, Missiles and Satellites	• Security systems • Telephone and banking • Defense and aerospace • Communication
Peripherals and Computer Networking	• Displays and monitors • Networking systems • Image processing • Network cards and printers
Consumer Electronics	• Digital cameras • Set top boxes • HD TVs • DVDs

- **Washing Machine** : It is an automatic clothes washing system. It contains Status display panel, switches and dials, motor, power supply and control unit, inner water level sensor and solenoid valve.
- **Automotive Electronics** : Electronic control units used in chassis, body electronics, security, power train, in-vehicle entertainment, and information systems.
- **Military / Aerospace** : Satellite systems, radar, sonar, navigation, weather systems, flight control systems, aircraft management systems.

□ Examples of embedded systems :

1. Personal digital assistant (PDA)
2. Printer
3. Cell phone
4. Automobile: engine, brakes, dash, etc.
5. Television
6. Household appliances.



Fig. 1.1.2



Fig. 1.1.3



Fig. 1.1.4



Fig. 1.1.5



Fig. 1.1.6



Fig. 1.1.7

📖 1.1.6 Purpose of Embedded System

□ 1. Data collection, storage and representation

- Purpose of embedded system design is data collection. It performs acquisition of data from the external world.
- Data collection is usually done for storage, analysis, manipulation and transmission.

- Data is a collection of facts, such as values or measurements. It can be numbers, words, measurements, observations or even just descriptions of things. Data can be analog or digital.
- Analog data is directly collected from the analog signal. There is no need to conversion of data.
- Embedded systems with analog data capturing techniques collect data directly in the form of analog signal whereas embedded systems with digital data collection mechanism convert the analog signal to the digital signal using analog to digital converters.

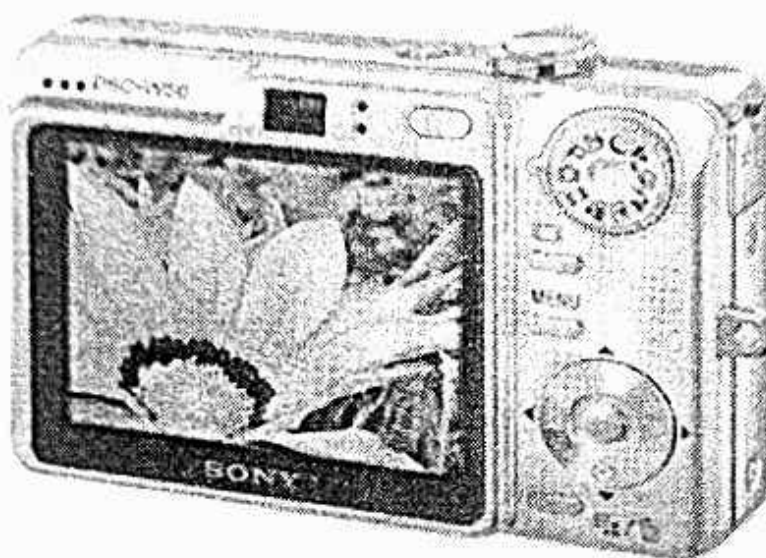
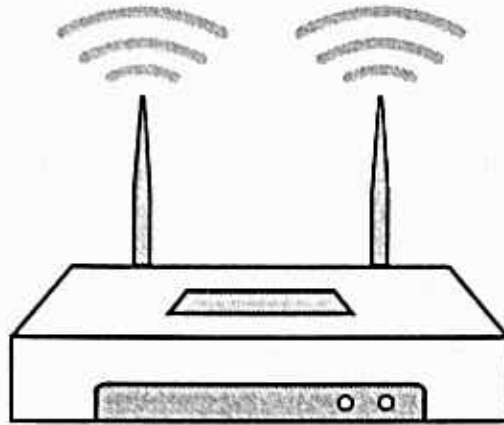


Fig. 1.1.8

- If the data is digital it can be directly captured by digital embedded system. Digital camera is example of embedded system.
- Collected data is directly stored in the system or directly processes on the data or it transmit to other system or deleted after processing.
- Images are captured and the captured image may be stored within the memory of the camera. The captured image can also be presented to the user through a graphic LCD unit.

❑ 2. Data communication

- Embedded data communication systems are in home networking systems to complex satellite communication application.
- Data transmission is in the form of wire medium or wireless medium. Data can either be transmitted by analog means or by digital means. Fig. 1.1.9 shows wireless router.

**Fig. 1.1.9**

- Bluetooth, ZigBee and Wi-Fi is an example of wireless communication and USB, TCP/IP is an example of wired communication.
- Initially wired medium is used in embedded system. As technology changes, wireless medium becomes de-facto standard in embedded system.
- Network hubs, routers, switches are examples of dedicated data transmission embedded systems.

☐ 3. Data processing

- Embedded systems with signal processing functionalities are employed in applications demanding signal processing like speech coding, audio video codec, transmission applications etc.
- A digital hearing aid is a typical example of an embedded system employing data processing.
- Digital hearing aid improves the hearing capacity of hearing impaired person.

☐ 4. Monitoring

- Medical domain embedded product come under monitoring function. Patient heartbeat is monitored by Electro cardiogram machine but it cannot impose control over the heartbeat.
- Digital CRO, digital multi-meters, and logic analyzers are the example of monitoring embedded devices.

☐ 5. Control

- In embedded system, sensor and actuators are used for controlling the system. Sensors are connected to input port. Actuators are connected to output port.

- Sensor captures the changes in the environmental variable. Middle system process the information. Actuators are changed according to the input variable. It display the output.
- Example of control is air conditioner system. It controls the room temperature to a specified limit.

❑ 6. Application-Specific Circuits (ASICs)

- For high-performance applications and for large markets, Application-Specific Integrated Circuits (ASICs) can be designed.
- However, the cost of designing and manufacturing such chips is quite high. Therefore, ASICs are appropriate only if either maximum energy efficiency is needed and if the market accepts the costs or if a large number of such systems can be sold.

1.1.7 Embedded System vs General Computing System

Embedded system	General computing system
It is combination of special purpose H/W and Embedded OS for executing a specific set of application	It is combination of generic hardware and general purpose OS for executing variety of Applications.
Not end-user programmable	End-user programmable
Key deciding factors is application specific requirements	Performance is the key deciding factor in the selection of the system
Response time requirement may or may not critical	Response requirements are not timer critical
Execution behaviour is deterministic	Exaction behaviour is not deterministic
Runs a few applications often known at design time	Intended to run a fully general set of applications
Differentiating features : Power, Cost, Size and Speed	Differentiating features : Speed, cost and software compatibility

Review Question

1. What are the different characteristics that an embedded system should possess ?

SPPU : May-18, Marks 5

1.2 Real Time Systems

- A real-time system is any information processing system which has to respond to externally generated input stimuli within a finite and specified period.
- Time constraints is the key parameter for real timesystems. It controls autonomous system such as robots, satellites, air traffic control and hydroelectric dams.
- **Determinism** : A system is deterministic if it always produces the same output for a known input. The output of a nondeterministic system will have random variations.
- **Deadline** : A deadline is the finite window of time in which a certain task must be completed.
- **Quality of Service** : The overall performance of a network. Includes factors such as bandwidth, throughput, availability, jitter, latency, and error rates.
- Real-Time system is defined as a system where the correctness of the system depends not only the result of computations but also on the time at which it is produced.
- When user gives an input to the system, it must process within the time limit and result is sent back. Real time system fails if it does not give result within the time limits.
- A real-time system is any information processing system which has to respond to externally generated input stimuli within a finite and specified period.
- A real time system is NOT a system that runs quickly. Real time system meets the temporal constraints. Examples of temporal constraints are as follows :
 - a. Few milliseconds for radar systems.
 - b. One second for machine-man interfaces (in an aircraft).
 - c. Hours for some chemical reactions.
 - d. 24 hours for weather forecast.
 - e. Several months or years for some spacecrafts
- **Real-time systems** are those systems in which the overall correctness of the system depends on both the functional correctness and the timing correctness. Real time systems also have a substantial knowledge of the system it controls and the applications running on it. These are deadline dependent.
- Real time systems are of two types : **Hard real time and soft real time.**
- Critical task is completed within the time limit in hard real time system. All the delay in the system is fixed and time bounded. Existing general purpose operating system does not support the hard real time system functions. Real time task cannot keep waiting for longer time without allocating kernel.

- Soft real time system is less restrictive type. Soft real time cannot guarantee that it will be able to meet deadline under all condition. Example of soft real time system is digital telephone and digital audio.
- Real time operating system uses priority scheduling algorithm to meet the response requirement of a real time application. General real time applications with their example are listed below :
 1. **Transportation** : Air traffic control and traffic light system
 2. **Communication** : Digital telephone
 3. **Process control** : Petroleum and paper mill
 4. **Detection** : Burglar system and radar system
 5. **Flight simulation** : Auto pilot shuttle mission simulator.
- Memory management in real time system is less demanding than in order type of multiprogramming operating systems. Time critical device management is one of the main characteristics of the real time systems.

1.2.1 Difference between Hard and Soft Real Time System

Sr. No.	Hard real time	Soft real time
1.	Hard response time is required	Soft response time is required.
2.	Data integrity is short term	Data integrity is long term
3.	Size of data file is small or medium	Size of data file is large
4.	Peak load performance is predictable	Peak load performance is degraded.
5.	Hard real-time systems have little laxity and generally provide full deadline compliance.	Soft real-time systems are more flexible. They have greater laxity and can tolerate certain amounts of deadline misses.
6.	Safety critical systems are typically hard real time systems	Linux and many OS provide soft real time system.

1.2.2 Real Time OS

- A Real-Time Operating System (RTOS) is an operating system intended to serve real-time application process data as it comes in, typically without buffering delays.
- In general, an operating system is responsible for managing the hardware resources of a computer and hosting applications that run on the computer. An RTOS performs these

tasks, but is also specially designed to run applications with very precise timing and a high degree of reliability. This can be especially important in measurement and automation systems where downtime is costly or a program delay could cause a safety hazard.

- An RTOS can guarantee that a program will run with very consistent timing. Real-time operating systems do this by providing programmers with a high degree of control over how tasks are prioritized, and typically also allow checking to make sure that important deadlines are met.
- Fig. 1.2.1 shows hardware and software hierarchy in RTOS.

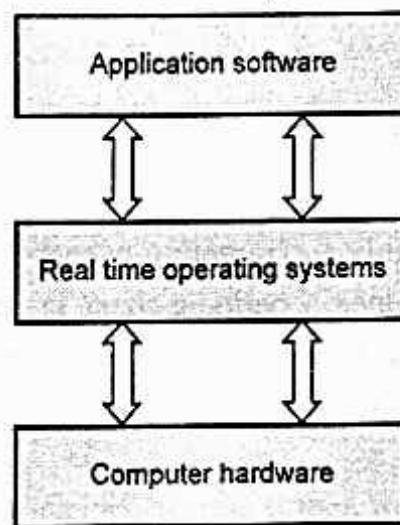


Fig. 1.2.1 : RTOS

- A real-time kernel is software that manages the time and resources of a microprocessor, microcontroller or Digital Signal Processor (DSP), and provides indispensable services to your applications.

□ RTOS kernel services :

- Along with objects, most kernels provide services that help developers create applications for real-time embedded systems.
 1. Timers
 2. Device I/O supervisor
 3. Dynamic memory allocation
 4. Intertask communication and synchronization
 5. Task management
- RTOS generally contains a real-time kernel and other higher-level services such as file management, protocol stacks, a graphical user interface and other components. Most additional services revolve around I/O devices.

- The core of an RTOS is known as the kernel. An API is provided to allow access to the kernel for the creation of threads, among other things. A thread is like a function that has its own stack, and a Thread Control Block (TCB). In addition to the stack, which is private to a thread, each thread control block holds information about the state of that thread.
- The kernel also contains a scheduler. The scheduler is responsible for executing threads in accordance with a scheduling mechanism. The main difference among schedulers is how they distribute execution time among the various threads they are managing. Priority-based, preemptive scheduling is the most popular and prevalent thread scheduling algorithm for embedded RTOSes. Typically, threads of the same priority execute in a round-robin fashion.
- Real-time operating systems normally support two main types of timers: periodic timers and aperiodic timers.
- **Periodic timers** are used mainly for sampling events at regular intervals or performing some activities periodically. Once a periodic timer is set, each time after it expires the corresponding handler routine is invoked, it gets reinserted into the timer queue. **Aperiodic timers** are set to expire only once. Watchdog timers are popular examples of one shot timers.
- Kernel objects are special constructs that are the building blocks for application development.
 1. **Tasks** : Concurrent and independent threads of execution that can compete for CPU execution time
 2. **Semaphores** : Token-like objects that can be incremented or decremented by tasks for synchronization or mutual exclusion
 3. **Message queues** : Buffer-like data structures that can be used for synchronization, mutual exclusion, and data exchange by passing messages between tasks

1.3 Real Time Tasks

- Task is a set of related jobs which jointly provide some function. A job is a unit of work that is scheduled and executed by a system.
- Real time tasks get generated in response to some events that may either be external or internal to the system. When task get generated, it is said to have arrived. Every real time system usually consists of a number of real time tasks.

- Job : A job is a unit of work scheduled and executed by the system.
- A task $T = \{ J_1, J_2, \dots, J_n \}$ is a set of related jobs that together perform some operation.
- Jobs execute on a processor and may depend on some resources. A scheduling algorithm describe how jobs execute.
- Processors are active devices on which jobs are scheduled. For example: threads scheduled on a CPU, data scheduled on a transmission link. A processor has a speed attribute, that determines the rate of progress of jobs executing on that processor.
- A resource (R) is a passive entity on which jobs may depend. Resources have different types or sizes, but have no speed attribute and are not consumed by use. Jobs compete for resources, and can block if a resource is in use

1.3.1 Clock Based Tasks

- Clock based tasks are also called **cyclic or periodic tasks**.
- All jobs of a periodic task τ_i have a regular inter arrival time T_i , we call τ_i the period of the periodic task τ_i .
- If a job for a periodic task τ_i arrives at time t , then the next job of task τ_i must arrive at $t + T_i$.
- Example of periodic task is task monitoring temperature of a patient. It is time driven method.
- Each task is repeated at a regular interval and maximum execution time is the same for each period.
- Arrival time is usually the start of the period and deadline is usually the end. Periodic tasks are mutually independent.
- Static scheduling is used for scheduling periodic tasks.

Plant time constant $\rightarrow$ Sampling time (T_s) $\rightarrow$ Interrupt

- The completion of the operations within the specified time is dependent on the number of operations to be performed and the speed of the computer.
- Synchronization is usually obtained by adding a clock to the computer system, and using a signal from this clock to interrupt the operation of the computer at predetermined fixed time interval.

1.3.2 Event Based Tasks

- Event based task is also called aperiodic task.
- Actions are to be performed not at particular times or time intervals but in response to some event. The system must respond within a given maximum time to a particular event.
- Events occur at non-deterministic intervals and event-based tasks are referred to as “a periodic” task.
- Aperiodic tasks is a stream of jobs arriving at irregular intervals. Each task can arrive at any time.
- These tasks are event driven. Example of this type of task is task activated upon detecting change in patients condition.
- Two aperiodic tasks can arrive at the same time.
- Dynamic scheduling is used to schedule both periodic and aperiodic tasks.
- The specification of event based systems usually indicates that the system must respond within specified maximum time to a particular event. These systems uses interrupts to indicate the real time system that the action is required.

1.3.3 Interactive Systems

- They represent the largest class of RTSs such as automatic bank tellers, reservation systems for hotels , airlines and car rental, etc.
- The real-time requirement is usually expressed in terms such as “the average response time must not exceed”
- The system responds at a time determined by the internal state of the computer and without any reference to the environment.
- The combination of clock based system and Event based system which gives the importance of average execution time of the task is called interactive systems.
- This covers the systems like automatic teller machine, reservation system for hotels, Airlines booking etc.
- This system receives the input from the plant or operator and initiates the task and executes within the average response time.
- For an example if you want draw cash from ATM when you put your card then it process the task of giving the money out. In this case the response time depends on the network traffic and internal processing time and it does not bother about other atmospheric changes.

1.4 Components of Embedded System

- Fig. 1.4.1 shows one possible organization for an embedded system.

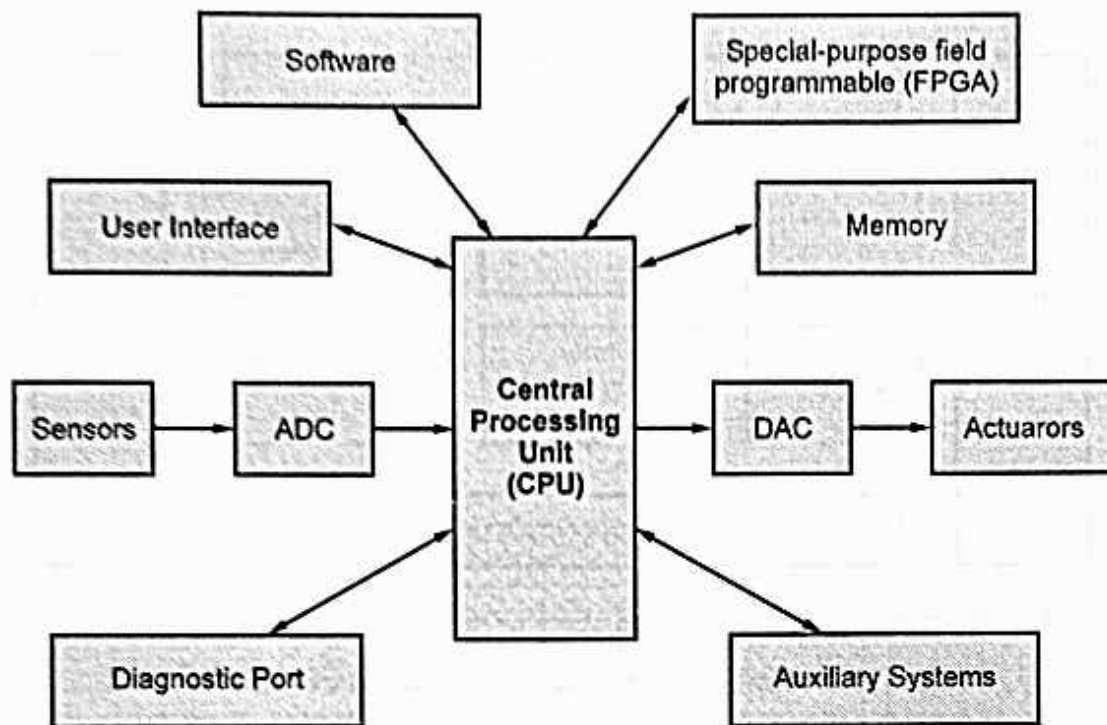


Fig. 1.4.1 : Possible organization for an embedded system

- An embedded system has three main components: Hardware, Software and time operating system.
- Hardware parts includes power supply, processor, memory, times and communication ports, system application specific circuit etc.
- Software parts includes the application software is required to perform the series of tasks. An embedded system has software designed to keep in view of three constraints :
 - Availability of system memory
 - Availability of processor speed
 - The need to limit power dissipation when running the system continuously in cycles of wait for events, run , stop and wake up.
- Real time operating system supervises the application software and provides a mechanism to let the processor run a process as per scheduling and do the switching from one task to another task.

1.4.1 Core Components of Embedded System

- Fig. 1.4.2 shows core of embedded system.

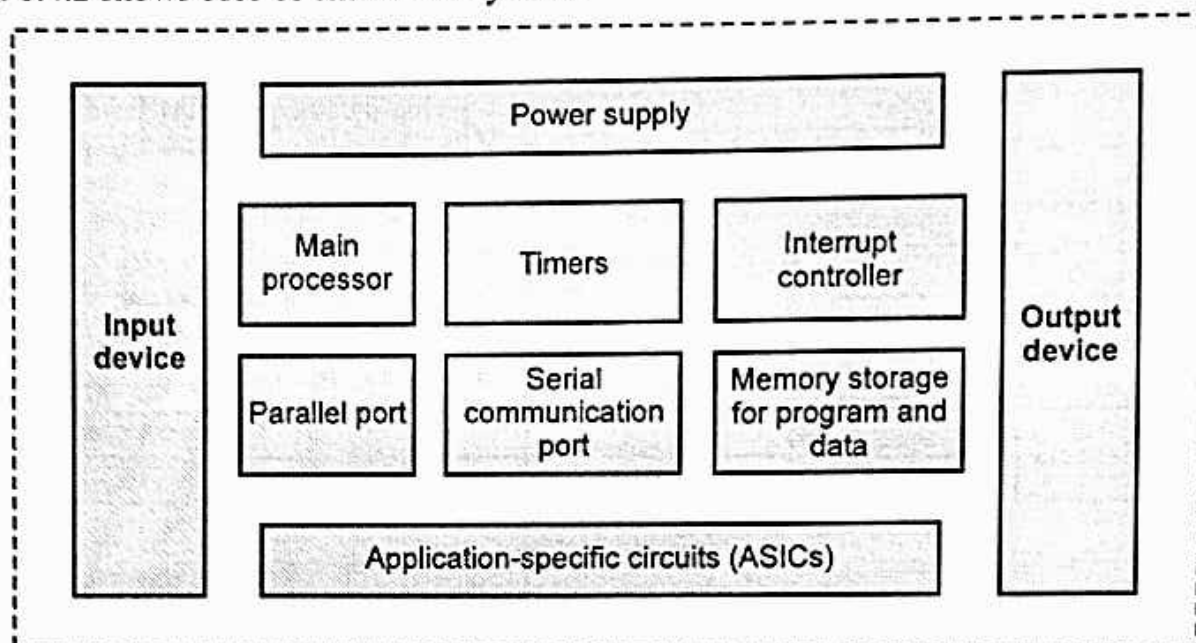


Fig. 1.4.2 : Embedded system hardware component

Components	Description	Merits	Demerits
Microprocessor	It is central processing unit and perform arithmetic and logical operations.	- General purpose is design and operation. - High processing power	- Does not contains built-in I/O port. - Dependent unit - Power saving options are less
Microcontroller	It is self contained unit and it does not require external interrupt controller, timer etc.	- Highly integrated chip - Many power saving features	Low processing power
Digital signal processor	Designed for performing high speed computational operations for addition, subtraction, division and multiplication	- Faster than general microprocessor - Used for real time processing	It performs signal processing according to stored program memory.

Components	Description	Merits	Demerits
ASIC	Designed to perform specific application	1. Consume small area 2. Reduces system development cost	1. Proprietary product 2. NRE cost is high
PLD	An IC that contains large numbers of gates, flip-flops, etc. that can be configured by the user to perform different functions is called a Programmable Logic Device.	1. It reduces total costs. 2. During design phase customer can change the circuitry. 3. No NRE cost	1. It may creates problem if customer can change design repeatedly.
COTS	COTS Product is a software product that has been sold, leased, or licensed to the general public by a commercial entity in a business of making a profit from product with multiple identical copies available	1. To reduce software life-cycle 2. To reduce development cost	1. Problems with COTS reuse include lack of control over functionality, performance, and evolution and problems with inter-operation

1.4.2 Microprocessor

- Embedded microprocessor typically does not have floating point or memory management units but often integrate various peripheral functions with the processor to reduce system cost. This makes them more application specific.
- Digital Signal Processors (DSP) are the most specialized embedded microprocessors. Designed for real time processing of digitized analog signals.
- The microprocessor, also known as the Central Processing Unit (CPU), is the brain of all computers and many household and electronic devices.
- CPU fetch and process a set of general purpose instruction. In most architectures it has three parts : an Arithmetic Logic Unit (ALU), a control unit and a set of registers, fast storage locations.

- **CPU clock** : The master coordinator of the CPU cycles, a conductor that synchronizes the CPU and its communication with other components.
- **Clock speed** : Number of cycles per second the CPU executes.
- The clock speed is measured in cycles per second, or hertz, such as 2 GHz, which is 2 billion cycles per second. Overall amount of work accomplished by the CPU depends on both the clock speed and the amount of computing accomplished on each cycle, which varies from CPU type to CPU type.
- CPU instruction set are as follows :
 - a. Data transfer operation
 - b. I/O operations
 - c. Stack operation
 - d. ALU operation
 - e. Program control and sequencing operation
- The CPU has two main components, namely :
 1. The Control Unit : It controls the fetching of instructions from the main memory and the subsequent execution of these instructions. Among other tasks carried out are the control of input and output devices and the passing of data to the arithmetic/logical unit for computation.
 2. The Arithmetic/Logical Unit (ALU) carries out arithmetic operations on integer and real operands. It can also perform simple logical tests for equality and greater than and less than between operands.
- Simple 8-bit and 16-bit based architectures are common, although the x86-compatible AMD's Athlon and Intel Atom are examples of 64-bit designs used in some relatively low power and low cost segments.

1.4.3 Microcontroller

- A microcontroller is a highly integrated chip which includes, on one chip, all or most of the parts needed for a controller. The microcontroller could be called a "one-chip solution". It typically includes CPU, RAM, ROM, I/O, timers and interrupt controller.
- Embedded controller may not be a microcontroller per se but is used for special purpose control application.
- Microcontrollers are normally integrated into small devices like the microwave ovens, keyboards and cell phones. It requires a small amount of computing power, less memory and very little attachment accessories.
- Example : Motorola's 6811, Intel's 8051, Zilog's Z8 and PIC 16X

- Intel 8051 microcontroller has the following characteristics :
 1. 8 bit CPU, optimized for control applications,
 2. Large set of operations on Boolean data types,
 3. Program address space of 64 kbytes,
 4. Separate data address space of 64 kbytes,
 5. 4 kbytes of program memory on chip, 128 bytes of data memory on chip,
 6. 32 I/O lines, each of which can be addressed individually,
 7. 2 counters on the chip,
 8. Universal asynchronous receiver/transmitter for serial lines available on the chip,
 9. Clock generation on the chip,
 10. Many variations commercially available.

1.4.4 Difference between Microcontroller and Microprocessor

Sr. No.	Microcontroller	Microprocessor
1.	CPU, RAM, ROM, I/O and timer are all on a single chip	CPU is stand-alone, RAM, ROM, I/O, timer are separate
2.	Fixed amount of on-chip ROM, RAM, I/O ports	Designer can decide on the amount of ROM, RAM and I/O ports.
3.	Low processing power	High processing power
4.	Low power consumption	High power consumption
5.	Instruction sets focus on control and bit-level operations	Instruction sets focus on processing-intensive operations
6.	Typically 8/16 bit	Typically 32/64 bit
7.	Typically single-cycle/two-stage pipeline	Typically deep pipeline
8.	Many type of bit handling instruction	Bit handling instruction is less, one or two type only
9.	They are used for designing application specific dedicated systems	It is used for designing general purpose digital computers system



1.4.5 Embedded Processor

- Embedded processor consists of program flow control unit and execution unit. The program flow control unit fetches an instruction from the memory. Execution unit perform data transfer operation and conversion of data from one format to another format.
- Execution unit contains Arithmetic and Logic Unit (ALU) and other circuit for executing instruction for program. It support instruction like jump, halt, interrupt, and branch and function call.
- Processor makes use of fetch-and-execute cycle. Processor instruction set defines an instruction and it is executed in sequence by processor. A processor is in the form of an IC chip.
- Embedded system processor chip/core is in the following form :
 1. General Purpose Processor (GPP)
 2. Application Specific Instruction Set Processor (ASIP)
 3. Multiprocessor
 4. GPP/ASIP core integrated into either ASIC or VSLI circuit.
- Important characteristic of embedded processors is their static structure. The end user has very limited access to programming. Most of the software is provided by the processor integrator and/or application developer, is resident on ROM memories, and runs on processors which are not visible to the end user. The user cannot change the basic actions of the processor, but it can program the processor to perform a sequence of basic actions.
- Embedded processors are essentially non-homogeneous processors and this characteristic is induced by the heterogeneous character of the process within which the processor is embedded.
- A typical processor is not only mixing hardware design with software design but also mixes design styles within each of these categories. The heterogeneous character can be seen in many aspects of the processors as follows :
 1. Both analog and digital sub-processors may be present in the system;
 2. The topology of the system is rather irregular
 3. The hardware may include microprocessors, micro-controllers, DSPs, ASICs, FPGAs, etc.
 4. The software may include various software modules as well as a multitasking real-time operating system.
- A large number of embedded processors perform real time processing which is the jobs executed by the processor deadlines. The deadlines can be hard and soft.

- **Hard real-time deadline :** Something really bad happens if the processor does not deliver its output before the deadline.
- **Soft real-time deadline :** Essential task, e.g., display update, is always completed even when its deadlines cannot be reached in time; non-essential task, e.g., connection establishment, may be aborted if it cannot be completed in time.
- Most embedded systems place severe constraints on the processor in the terms of requirements for size, weight, power, cooling, performance, cost and reliability. This is because the processor is just a component of a larger system which has its own operating requirements and manufacturing constraints.



1.4.6 Digital Signal Processor

- Digital Signal Processing (DSP) manipulates different types of signals with the intention of filtering, measuring, or compressing and producing analog signals.
- DSP is special purpose microprocessor and designed to meet the computational demands and power constraints of today's embedded application.
- Analog signals differ by taking information and translating it into electric pulses of varying amplitude, whereas digital signal information is translated into binary format where each bit of data is represented by two distinguishable amplitudes.
- Another noticeable difference is that analog signals can be represented as sine waves and digital signals are represented as square waves.
- Hardware is used for implementing algorithm in DSP. So it speed up the execution. General purpose processor implement algorithm is firmware so speed of execution depends on the clock for the processor.
- DSP contains 4 key components :
 1. **Computing engine :** Mathematical manipulations, calculations, and processes by accessing the program, or task from the program memory and the information stored in the Data Memory.
 2. **Data memory :** This stores the information to be processed and works hand in hand with program memory.
 3. **Program memory :** This stores the programs, or tasks, that the DSP will use to process, compress, or manipulate data.
 4. **I/O :** This can be used for various things, depending on the field DSP is being used for, i.e. external ports, serial ports, timers and connecting to the outside world.
- Digital signal processing uses a large amount of real time calculations. DSP used in multimedia application, telecommunication and audio-video signal processing.

1.4.7 Application Specific Integrated Circuit

- In Integrated Circuit (IC) designed to perform a specific function for a specific application. Most of the ASIC are proprietary product.
- Here a complete system or product is integrated onto a chip and virtually no other components are required. The cost of designing an ASIC is very high and therefore they tend to be reserved for high volume products. But ASICs can be very cost effective for many applications where they are produced in large scale.
- ASIC technology permits a designer to tailor a design to meet specific requirements. For example, circuit boards can be fabricated to fit into existing modules and racks, and function scan be distributed to minimize rewiring.
- In a full custom ASIC an engineer designs some or all parts of the logic cells, circuits or layouts. A microprocessor is an example of full-custom IC, gives the highest performance but the longest designing time and is very expensive, used to design very high performance systems.
- Initial investment is large for ASIC. Fabrication of ASICs requires a non-refundable initial investment for the process technology and configuration expenses. This type of investment is known as Non-Recurring Engineering (NRE) cost.
- If third party is ready to pay NRE cost and ASIC made available into the market the ASIC is referred as Application Specific Standard Product (ASSP).
- Features of ASICs
 1. Low (NRE) cost.
 2. Low power consumption.
 3. Less complex.
 4. High performance.
- There are some drawbacks of ASICs :
 1. Inflexible design
 2. Deployed systems cannot be upgraded.
 3. Mistakes in product development are costly.
 4. Updates requires a redesign.
 5. Complex & expensive development tool.

- Difference between ASIC and ASSP

Sr. No.	ASIC	ASSP
1.	It is microchip designed to perform specific application	If third party is ready to pay NRE cost and ASIC made available into the market the ASIC is referred as Application Specific Standard Product (ASSP).
2.	Most of the ASIC are proprietary product.	Openly available in the market

1.4.8 Programmable Logic Device

- An IC that contains large numbers of gates, flip-flops, etc. that can be configured by the user to perform different functions is called a Programmable Logic Device. The main reason for using programmable logic is to reduce total costs.
- Compared to ASICs, PLDs offer lower NRE costs, fast design turn-around, lower risk and simpler design tools.
- The term "programmable" means changing either hardware or software configuration of an internal logic and interconnects.
- Logic devices are divided into two types : Fixed and programmable.
- **Fixed Logic Devices** : Circuit is permanent and perform one function or set of functions. Once it is manufactured, it cannot change.
- **Programmable** : Device can be reconfigured to perform any number of functions. DVD player, router and DSL modem is an example of this type.
- Types of PLD are :
 1. Complex Programmable Logic Device
 2. Field-Programmable Gate Array.

Complex Programmable Logic Device (CPLD)

- A CPLD has a few PLD blocks or macro cells on a single device or chip.
- These blocks are interconnected with some programmable connections in between. Each PAL-like block is also connected to I/O block.
- Thus a CPLD has two levels of programmability : (i) Each PLD block can be programmed and (ii) The interconnections between the PLDs can also be programmed.
- As with PALs, CPLDs have an internal configuration memory and are also programmed with device programmers.

- The advantage of CPLDs is that more complex designs can be implemented. Propagation time is only slightly worse than an equivalent-process PAL due to one extra set of programmable interconnections.

Field-Programmable Gate Array

- FPGA is an array of Configurable Logic Blocks (CLBs) surrounded by programmable input and output blocks.
- FPGA, like the gate array, places fixed cells on the wafer and the FPGA designer constructs more complex functions from these cells.
- Different FPGA manufacturers have developed different basic cells, seeking to provide the most useful functionality in the cells for generation of overall FPGA functions.
- Cells range from fine-grained cells consisting of basic gates through medium-grained cells providing more complex programmable functions to large-grained cells.
- A typical FPGA is the Xilinx XC4003. The XC4000 family is a coarse-grained architecture in which each CLB has 2 sets of 4 inputs, a second level of logic to combine the two intermediate outputs and 2 registered outputs.
- One disadvantage of FPGAs is the relatively large propagation delays. Another disadvantage of FPGAs is the expensive and slow design software.

1.4.8.1 Comparison of CPLD and FPGA

Sr. No.	CPLD	FPGA
1.	PLD is used for construction of CPLD.	Logic blocks are used for construction of FPGA.
2.	It is non-volatile.	It is volatile.
3.	Delays are much more predictable in CPLDs.	Prediction of delay is difficult in FPGA.
4.	CPLDs are less costly.	FPGAs are costly.
5.	Suited for control circuit.	Suited for timing circuit.
6.	Operating speed is slow.	Operating speed is high.
7.	CPLD has less flexibility and design capacity.	FPGA has more flexibility as well as design capacity.
8.	The CPLD could work immediately after power up.	FPGA could not work until the configuration is done.
9.	CPLDs are 'coarse-grain' devices	FPGAs are considered as 'fine-grain' devices.

1.4.8.2 Difference between FPGA and ASIC

Sr. No.	FPGA	ASIC
1.	An FPGA is a reprogrammable integrated circuit.	An ASIC is a unique type of integrated circuit meant for a specific application.
2.	FPGA is not efficient in terms of use of materials. A certain number of components are always wasted.	An ASIC wastes very little material, recurring costs are low.
3.	FPGA is better than an ASIC when building low volume production circuits.	Cost of ASIC is low only when it is produced in large quantity.
4.	FPGA is alterable.	An ASIC can no longer be altered once created.
5.	FPGAs are useful for research and development activities. Prototype fabrication using FPGA is affordable and fast.	ASICs are not suitable for research and development purposes, as they are not reconfigurable.

1.4.9 Commercial Off the Shelf Components

- COTS is traditional methodology for Embedded Systems. It minimizes hardware costs and increases design & verification costs.
- COTS Product is a software product that has been sold, leased, or licensed to the general public by a commercial entity in a business of making a profit from product with multiple identical copies available
- Commercial Off-The-Shelf systems (COTS) systems are usually complete application systems that offer an Application Programming Interface. Building large systems by integrating COTS systems is now a viable development strategy for some types of system such as E-commerce systems.
- The key benefit is faster application development and usually, lower development costs.
- COTS solutions tend to have a shorter prototyping phase, since the hardware is ready-to-use out of the box and troubleshooting involves only rewiring, reprogramming, or changing the types and quantities of COTS I/O modules. Hence, most of the effort is in software development and the NRE in prototyping will be smaller than for the custom approach
- Remote control toy car is an example of Commercial Off-The-Shelf systems.
- Problems with COTS reuse include lack of control over functionality, performance and evolution and problems with inter - operation.

- If the COTS hardware needs to be reworked on a different platform to build the end-product, you will spend significant NRE on the end-product design and development.
- Benefits of COTS-based system :
 1. To Improve software development process
 2. To reduce coding, debugging, unit testing, code inspection
 3. To reduce software life-cycle
 4. To reduce development cost

1.5 System on Chip and its Example

**SPPU : May-19**

- The semiconductor industry has continued to make impressive improvements in the achievable density of very large-scale integrated (VLSI) circuits. In order to keep pace with the levels of integration available, design engineers have developed new methodologies and techniques to manage the increased complexity inherent in these large chips.
- One such emerging methodology is system-on-chip(SoC) design, wherein predesigned and pre-verified blocks often called intellectual property (IP) blocks, IP cores, or virtual components are obtained from internal sources, or third parties, and combined on a single chip.
- These reusable IP cores may include embedded processors, memory blocks, interface blocks, analog blocks and components that handle application specific processing functions. Corresponding software components are also provided in a reusable form and may include real-time operating systems and kernels, library functions, and de-vice drivers.
- The main prerequisite for creating SoC designs is a set of reusable IP blocks that support plug-and-play integration. IP blocks are the highest level building blocks of an SoC.
- A library of reusable IP blocks with various timing, area, power configurations is the key to SoC success as it allows mix-and-match of different IP blocks so that the SoC integrator can apply the tradeoffs that best suit the needs of the target application. The process of creating reusable IP block, however, differs from the traditional ASIC design approach.
- Demand for low cost and higher density platform requires the integration of devices. As integration levels increases, more and more logic is added to the processor die, creating families of applications specific service processors.
- System-on-Chip (SoC) designs increasingly become the driving force of a number of modern electronics systems. A number of key technologies integrate together in forming the highly complex embedded platform.

- SoC architects to employ a broad system level design strategy that will allow :
 1. Explore and evaluate system level architectural choices
 2. Concurrent hardware-software design
 3. Easily evaluate and integrate a number of different technologies
 4. Adequate verification at every level of the design cycle
 5. Carry out an architecture level power analysis
 6. Drive requirements for executable specifications
 7. Provide visibility into designs
- Conceptually System on Chip refers to integrating the components of a board onto a single chip. Looks straightforward but productivity levels are too low to make it a reality.
- Motivation for SoC Design :
 1. Higher productivity levels
 2. Lower overall cost
 3. Lower overall power
 4. Smaller form factor
 5. Higher integration levels
 6. Rapid development of derivative designs
- The SoC means chip, software and integration.

$\text{SoC} = \text{Chip} + \text{Software} + \text{Integration}$

1.5.1 Block of SoC

- Fig. 1.5.1 shows block diagram of SoC.

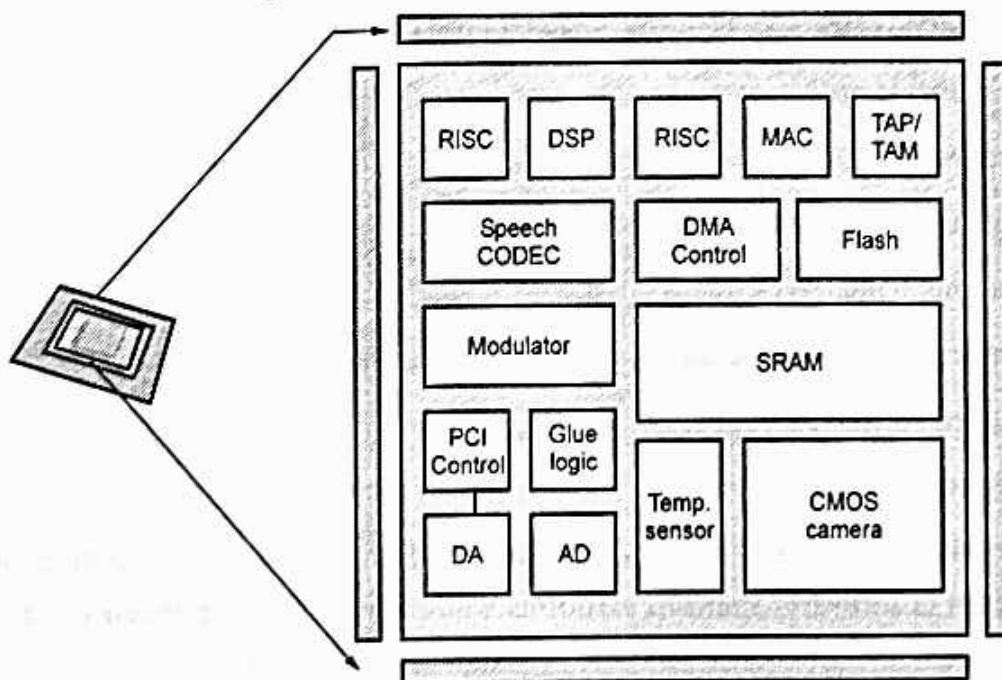


Fig. 1.5.1 : Block diagram of SoC

- The SoC chip includes :
 1. Embedded processor
 2. ASIC logics and analog circuitry
 3. Embedded memory
- The SoC software includes : OS, compiler, simulator, firmware; driver, protocol stack integrated development environment (debugger, linker, ICE) application interface (C/C++, assembly)
- The SoC integration includes :
 1. The whole system solution
 2. Manufacture consultant
 3. Technical supporting
- SoC consists of at least two or more complex micro-electronic macro components previously integrated into different single dies. Complex functionalities that previously required heterogeneous components to be connected on a PCB are integrated within one single silicon chip.
- Technologies implementing embedded systems evolved from micro-controllers and discrete components to fully integrated SoC. Reason is that a advances in Silicon process technology enabling a complete system to be designed into one or few integrated devices.
- SoC design is defined by extensive use of reusable IP blocks and mixed hardware and software design issues :
 1. Programmable processor
 2. Embedded memory
 3. Digital signal processors
 4. System bus + interfaces
 5. Embedded programmable logic
 6. Embedded software
 7. Analog components.
- Hardware is not sufficient to build an SoC platform. Most of the system on chip is implemented in software. Standard platforms with software differentiation will be the trend in the market.

1.5.2 Components of SoC

- SoC design is defined by extensive use of reusable IP blocks and mixed hardware and software design issues :
 1. Programmable processor
 2. Embedded memory
 3. Digital signal processors
 4. System bus + interfaces
 5. Embedded programmable logic
 6. Embedded software
 7. Analog components.....
- Fig. 1.5.2 shows block diagram with components of SoC.

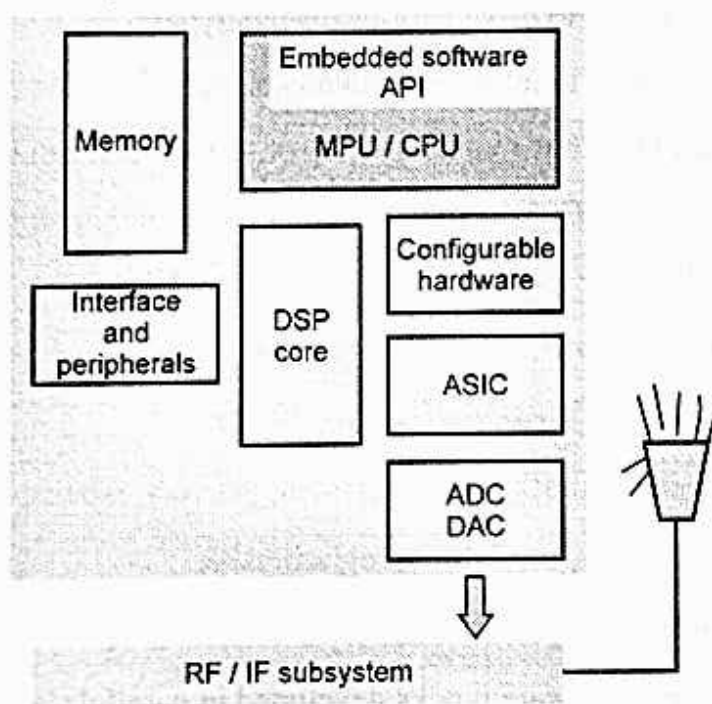


Fig. 1.5.2 : System on chip

- Hardware is not sufficient to build an SoC platform. Most of the system on chip is implemented in software. Standard platforms with software differentiation will be the trend in the market.
- Example : in set-top box market, manufactures are converting from 7 different chips to 1 chip with 7 different applications software.

- A typical SoC will contain :

1. A processor
2. Onboard execution memory (SRAM)
3. Peripheral systems and interfaces connected to the processing core via a SoC bus.
4. Many microcontrollers may contain FLASH memory for program storage.
1. **Memory controller** : Interfaces with the onboard RAM.
2. **DMA** : Handles the automated transfers of data between the RAM and memory-mapped hardware.
3. **USB controller** : Manages the hardware side of the device's USB connections.
4. **DSP core** : It provides hardware acceleration for some signal processing, such as JPEG encoding.
5. **Display** : Enables the SoC to drive various display types.
6. **Camera** : Allows the SoC to interface with a camera.
7. **Storage** : Manages I/O with the various types of storage that can be used with the SoC.
8. **Debug** : Enables the SoC to be connected to hardware debugging tools through various mechanisms, such as JTAG.

1.5.3 SoC Implementation

- Following methods are used for implementing SoC :
 1. ASIC - Application Specific IC, very integrated, yet very expensive
 2. FPGA - Cheaper to implement, field reprogrammable
 3. Programmable devices - Off the shelf devices, quick to program, cheap
 4. IP hardware blocks and software blocks developed in parallel
 5. Hardware in CAD
 6. Software in development environment
 7. Emulated and verified on FPGA
 8. Place and route, created in silicon
- On solution to the design productivity gap is to make ASIC designs more standardized by reusing segments of previously manufactured chips. These segments are known as "blocks", "macros", "cores" or "cells".

- The blocks can either be developed in-house or licensed from an IP company cores are the basic building blocks.

1.5.4 Example

- Embedded systems often have stringent performance and power requirements, leading designers to incorporate special-purpose hardware into their designs.
- Hardware-based implementations avoid the instruction fetch/decode/execute overhead of traditional software execution, and use resources spatially to increase parallelism.
- In many embedded applications, such as multimedia, encryption, wireless communication, and others, highly repetitive parallel computations well-suited to hardware implementation represent a significant fraction of the overall computation required by the system.
- Unfortunately, application-specific integrated circuit (ASIC) implementation is not feasible or desirable for all circuits. One key problem is that the non-recurring engineering costs (NREs) of ASICs have been increasing dramatically.
- Fig. 1.5.3 shows example of digital camera.

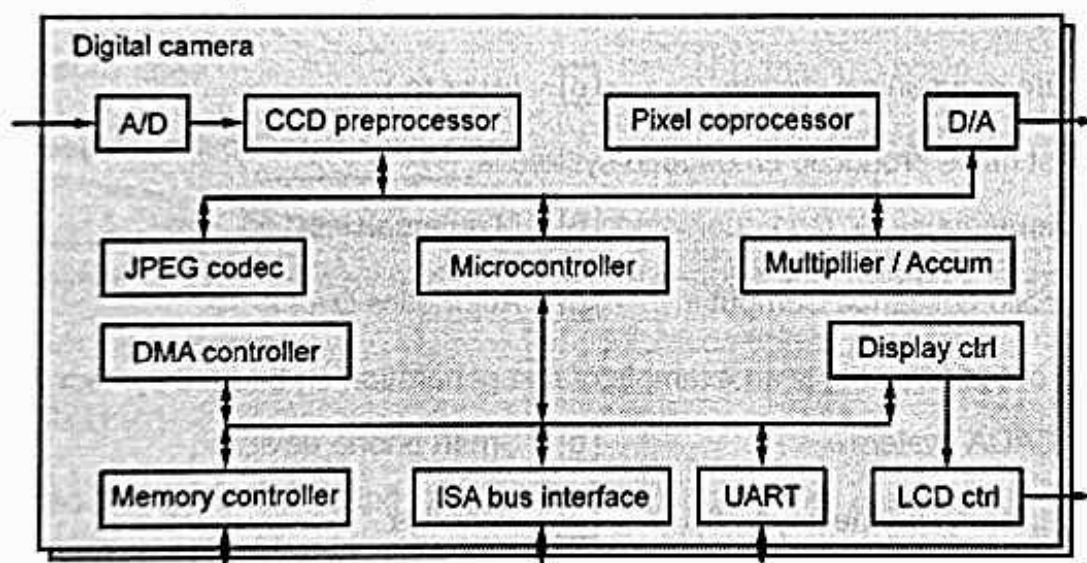


Fig. 1.5.3 : Design of digital camera

- The A/D and D/A circuits convert analog images to digital and digital to analog, respectively. The CCD preprocessor is a charge-coupled device preprocessor.
- The JPEG codec compresses and decompresses an image using the JPEG compression standard, enabling compact storage in the limited memory of the camera.
- The Pixel coprocessor aids in rapidly displaying images. The Memory controller controls access to a memory chip also found in the camera, while the DMA controller enables direct memory access without requiring the use of the microcontroller.

- The UART enables communication with a PC's serial port for uploading video frames, while the ISA bus interface enables a faster connection with a PC's ISA bus.
- The LCD ctrl and Display ctrl circuits control the display of images on the camera's liquid-crystal display device. A Multiplier circuit assists with certain digital signal processing. At the heart of the system is a microcontroller, which is a processor that controls the activities of all the other circuits.
- We can think of each device as a processor designed for a particular task, while the microcontroller is a more general processor designed for general tasks.

Review Question

1. What is SoC ? Explain major applications of SoC and benefits of it.

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